



Mathematical & Computational Sciences
UNIVERSITY OF TORONTO
MISSISSAUGA

STA304H5F
Surveys, Sampling and Observational Data
COURSE PROJECT - Fall 2024

Final Due Date: 02-12-2024

Project Guidelines Quiz	Form Groups	Proposal	Data Collection	Report (Rough Draft)	Report (Final Draft)
1%	1%	3%	3%	2%	10%

The TA, Anna Ly (annahuynh.ly@mail.utoronto.ca) will oversee the group projects for this semester.

- If the question is general, ask on Piazza.
- If the question is personal, ask via email.
- The TA will not read-over projects before submission through email. They will oversee them during booked office hour appointments conducted online on Zoom. You may book them through Google Sheets. The link will be available on Quercus at the start of the term.
 - Rules for booking:
 - Enter your group name, and number of members joining.
 - Don't erase other group's timings. If you do, your group will be **BANNED** from booking appointments with the TA.
 - If you made a mistake, email the TA immediately (before the other group notices) and you will be forgiven.
 - If someone erased your time, email me immediately and we will make up for another time at your convenience (I must see proof in the editing history.)

- You may only have 1 booking per checkpoint. If I see a group twice per checkpoint, then your group will be **BANNED** from booking future appointments.
 - Checkpoints are: Signatures/Group Definition, Proposal, Progress Report (Data), Technical Report (Rough), Technical Report (Final)
 - If students are caught misusing the sign-up sheet, the TA will move to an email-booking basis.
- The other TAs oversee facilitating tutorials & marking for examinations. Please do not ask them inquiries regarding the course project.
- Teaching Assistants are NOT permitted to privately tutor students for courses they are currently a teaching assistant for. Contacting us for private information, such as our Discord, Messenger, WeChat, etc., is **strictly prohibited**.

Project Policies

- This project is group work (except for filling out surveys). You can discuss with your group members only. It is plagiarism (a serious academic offence) to submit solutions in other people's words (including but not limited to other students, the instructor's, solutions from previous years or courses, websites, etc.) You may not use past projects considered by others; your group should complete the project on your own. You are responsible for knowing and adhering to the University of Toronto's Code of Behaviour on Academic Matters (see course outline).
- You are expected to use the programming language R, and any R library that exists online. You may NOT use any other programming language or software, such as Excel (besides formatting the codebook) or Python. **The use of other programming languages will result in a 20% reduction.**
- Regarding generative artificial intelligence tools, the use of such tools or apps, including ChatGPT or any other AI writing or coding assistants, is permitted under careful conditions. Students are expected to send screenshots or a link to their conversation with AI writing tools for each submission.
- Each member of the group should provide an equal contribution to the project. However, it is up to you and your group to decide how to allocate duties fairly. You all

do not have to be involved in each aspect. For example, you may not be able to meet and collect the data together.

- **Students within the same group will get the same mark**, except for the project guidelines quiz and filling out surveys. However, you may receive lower marks if we received emails from most of your group members saying that you did not contribute/cooperate with them. Thus, it is recommended that you save all proof of your group meetings (schedules/emails/chats/shared documents). You may be asked to provide it if there is a dispute about a particular group. For example, suppose there is a case where every group member confirmed that a particular student did not do any work. The other group members will have to show proof of the student's lack of participation and the student will have to show that they did do work. The course instructor will make a final decision. In rare cases where a group member does not do any work (and disappears from the project), that student will receive a 0 on the whole project if all other group members prove it.
- Groups experiencing significant internal conflicts or suspected of using excessive generative AI will be required to complete an oral examination to supplement their Report (Final Draft). **Details regarding the oral examination will be provided in a separate document.**

Submissions & Feedback

- For forming student groups (the first checkpoint), the submission will be through email by ONE representative.
- Once you are formed in your appropriate groups (the second checkpoint and beyond), you cannot send submissions through email. **Submissions through email beyond the first checkpoint will result in a 20% reduction.**
- Project's submissions should be uploaded on Quercus and are due by **11:59 pm EST** on the due date.
- **Late submissions will result in a 25% reduction per day.**
- You'll notice that you will be in a STA304 group after the first checkpoint. In your group page, there will be a "file" section. **You now must upload under the "file" section.** Any member of the group can submit on behalf of the entire group. You'll find that this is a convenient way to submit your reports for the following reasons:

- Unlimited re-submissions before the deadline.
 - Very easy to access and preview the file so you can see in case it is corrupted.
 - You can upload multiple files (this is useful for future checkpoints as you may want to upload extra files, such as new .CSV files, .R codes, .tex or .rmd files, and AI chatlog history).
 - The file name should correspond to the checkpoint.
 - You are allowed to create sub-folders for general organization.
 - Feedback (such as marked rubrics) will also be provided under the “files” section.
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Remarks Requests

- Remark requests must be sent via email to the TA. You may only submit a remark request once the TA has released marks for everyone, which will be announced.
 - You must add “STA304 Remark Request” in the subject line. It is your responsibility to defend your mark. Include the following in the body of the email:
 - Group number/name.
 - The reason why marks were deducted. Provide a screenshot in the commented feedback where the TA deducted marks.
 - The reason why you deserve these marks. Provide a screenshot of the portion of your project where you fulfilled the requirement and your justification.
 - Note that we can see the time when files were modified. If your file was modified AFTER the deadline, we will not remark the project.
 - **Note that it is possible your mark may lower or stay the same upon a remark request, or reported for AO, upon further analysis.**
 - If by chance the wrong marks were submitted, you only need to include the group number, lecture section, and what mark was written on the rubric.
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Reporting Inactive Members

We recommend for all group members to do all the following to keep as much evidence as needed to prove participation and lack thereof:

- Group chat history. (We have personally noticed Discord to be the easiest to organize since we can search the conversation history per user and explicitly see timestamps easily.)
- In your chat history, if it's the case that students have a particular part of the project, please include their roles and which checkpoints they were responsible for.
- Emails/shared documents.
- Screenshots of group meetings and attendance.
- Use tools that require login for edits, such as Google Drive/Docs, Overleaf, etc. Ensure you are logged in while making edits. This is so we can see the changes made per user in the revision history.

You must report all evidence within 24 hours of the due date of each checkpoint.

You also must do this at each checkpoint, even if the culprits have a previous history of not doing anything.

You must email using the following:

- Write to the TA and the course instructor.
- CC group members who have performed the work. We will need them to verify in the email chain that certain students have not done the work.
- In the subject title, include "STA304 Inactive Group Members | <LECTURE SECTION> | <GROUP NUMBER/NAME>"
- In the body, include the inactive student's name as well as much evidence that you can possibly provide (as given earlier.)

The course instructor (not the TA) will decide the appropriate punishment for the inactive student depending on the amount of evidence and the severity. In rare cases where a group member does not do any work and disappears from the project, the student will receive 0 on the project.

Brief Description of Checkpoints

Project Guidelines Quiz – Due Sept. 16 [1%]

- This quiz is designed to ensure students have read the project description and understand academic integrity policies.
 - It is meant to be stress-free as the quiz is open book with no time limit, and you have three attempts to complete the quiz with the highest score to be considered.
 - This is possibly one of the easiest quizzes you'll take as an undergraduate. Please do not discuss answers or questions amongst friends; they can just read this document.
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Group Definition – Due Sep. 16 [1%]

- Each group should have a minimum of 4 and a maximum of 5 students.
 - Groups of up to 6 members will be entertained on a case-by-case basis. Please email the TA if you want more members.
- To facilitate finding group members, there is a post on the Piazza Discussion Board to find team members.
- One member of each group is required to submit the names of all group members, by email, to the TA. Use the “Group Form” on Quercus and fill out the form entirely.
- You may resubmit the form as many times up until the deadline, in case you want to add or change members. If you want to exceed 5 members, you will be given a different form to fill out.
- If you cannot find your own group before the deadline, and you want to participate in the group project, you must email the TA two days prior to the deadline.

- If you have a couple of friends within the course, but struggle to find your final members, two days before the deadline you may email the TA using an incomplete version of the Group Forms to claim a topic prior to two days before the deadline.
 - Groups assigned by the TA will be given a three-day extension to submit the form.
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Proposal – Due Sep. 23 [3%]

- Students are expected to outline their plans for research and data collection. Details regarding the proposal are provided in a different document.
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Data Collection – Due Oct. 21 [3%]

- Students are expected to fill out at least 10 other surveys, as well as collect an appropriate amount of data, clean the data, and submit the codebook.
 - Marks for completing surveys will be individual and are worth 1%. As a safety measure, we recommend filling out more than 10 surveys in case another group fails to provide their dataset.
 - Details regarding data collection are provided in a different document.
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Technical Report – Rough Due Nov. 11 [2%], Final Due Dec. 2 [10%]

- The technical report should be treated almost as if you were writing a journal article; you might have read some of these during your undergraduate degree. These include an abstract, introduction, methodology, analysis, discussion, limitations, conclusion, and an appendix.
- We are expecting students to use either LaTeX, R-Markdown, or Quarto for the report.
- The goal for the group project is for students to learn. We have also noticed that students' writing abilities have deteriorated over the years and want students to have a report that they are proud to show in interviews. As a result, we implemented a low-stakes assessment as a chance for students to write-up what they perceive to be of decent quality, and then have the TA provide specific feedback before marking the final version.

- Do not take the “rough draft” checkpoint lightly. We recommend putting in as much effort as possible for you to receive as much feedback as possible. You may also add questions in between the rough draft if you had questions for us to answer via the feedback.
- As a result of implementing a rough draft, our standards for the final draft will be higher compared to reports that were possibly written by your peers that have taken this course in prior years. Do not rely on past examples to determine the quality necessary for full marks.
- Details regarding the technical report are provided in a different document.

Warning: groups that are suspected to have used an unauthorized amount of generative AI or with massive internal conflicts will need to conduct an oral examination, which will replace the technical report final draft mark. **Details regarding the oral examination is provided in a different document.**

Project Suggestions

Note that these are topic suggestions, not necessarily the title of your project. You are expected to slightly alter them to fit your case. However, once your idea is reported, you must stick to it up until the proposal. **Changing your project afterwards will result in a deduction of marks.** To avoid overlap, the TA will post a Piazza mega thread that showcases the taken topics once each group is formed.

1. Assessing Campus Dining Preferences.
2. Understanding Students' Study Habits and Academic Performance.
3. Student Perceptions of Campus Environmental Sustainability Initiatives.
4. Determining Student Preferences for Recreational Activities on Campus.
5. Proportion of Students Engaged in Extracurricular Activities.
6. Analyzing the Impact of Scholarship Programs on Student Retention.
7. Exploring the Relationship Between Faculty Mentorship and Student Achievement.
8. Student Preferences for Course Feedback Methods.
9. Analyzing Factors Affecting Student Commuting Choices.
10. Analyzing Student Participation in Campus Events.

11. Analyzing Factors Influencing Student Participation in Study Abroad Programs.
12. Assessing Student Views on Campus Security and Safety Measures.
13. Proportion of Students Utilizing Campus Gym and Wellness Centers.
14. Assessing Student Participation in Volunteer Activities.
15. Exploring the Relationship Between Nutrition and Academic Success.
16. Investigating the Impact of Social Media on Student Well-being.
17. Exploring Student Opinions on Course Registration Processes.
18. Analyzing Factors Affecting Student Preferences for Library Resources.
19. Exploring Student Preferences for Different Types of Learning Materials (e.g., videos, textbooks, generative AI such as ChatGPT).
20. Evaluating the Impact of Short Study Breaks on Concentration and Focus.
21. Exploring Student Opinions on Campus Residential Life and Amenities.
22. Exploring Student Opinions on Examination Formats and Effectiveness.
23. Exploring the Relationship Between Financial Stress and Academic Performance.
24. Assessing Student Preferences for Healthy Food Options on Campus.
25. Determining Student Preferences for Campus Parking Solutions.
26. Exploring the Relationship Between Classroom Environment and Student Engagement.
27. Evaluating the Use of E-books vs. Traditional Textbooks.
28. Exploring Student Preferences for Online vs. In-person Office Hours with Professors/TAs.
29. Measuring the Effectiveness of Online Collaboration Tools in Group Projects.
30. Evaluating the Impact of Part-Time Jobs on Student Academic Stress.
31. Exploring the Relationship Between Student Participation in Sports and Academic Performance.
32. Investigating the Relationship Between Campus Transportation and Academic Attendance.
33. Analyzing Factors Affecting Student Satisfaction with Campus Life.
34. Student Perceptions of Health and Wellness Resources on Campus.
35. Investigating the Impact of COVID-19 on Student Learning and Adaptation.
36. Assessing the Role of Campus Media in Shaping Student Opinions.
37. Analyzing Factors Influencing Student Choices in Graduate Programs.

- 38. Understanding the Impact of Academic Workload on Student Mental Health.
- 39. Exploring Student Preferences for Online Learning Platforms.
- 40. Assessing Student Preferences for Different Types of Learning Assessments.
- 41. Investigating the Impact of Office Hours on Student Learning.
- 42. Exploring the Relationship Between Social Media Usage and Academic Performance.
- 43. Analyzing Factors Influencing Student Participation in Online Discussion Forums.
- 44. Exploring the Relationship Between Faculty Teaching Styles and Student Learning Outcomes.
- 45. **Pick a Topic of Personal Interest. Something You Care About OR Something of Interest to Others!**